

E.E. 3333-001

Project 1

Remote Event Timer

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## Abstract

The purpose of this paper is to describe in detail the design and operation of a remote event sensing system that has been built as a Texas Tech project lab project. The system is essentially a timer that is activated or deactivated when an event at one of its transmitters occurs. The timer is accurate to 30+mins. Multiple events can be tracked by using an integrated RFID tag reader system and RFID tags with ID numbers loaded on them. The system implements a power amplifier to increase the usable range of the transmitters.

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## Introduction

The subject of this paper is a system that was developed to measure the time between two remote events. It contains two remote transmitters and a central receiver. Events occurring at either of the transmitters will activate or deactivate a timer within the central receiver. The timer must be accurate to the second for at least 15 minutes. The following figure contains an overview of the system [4]:

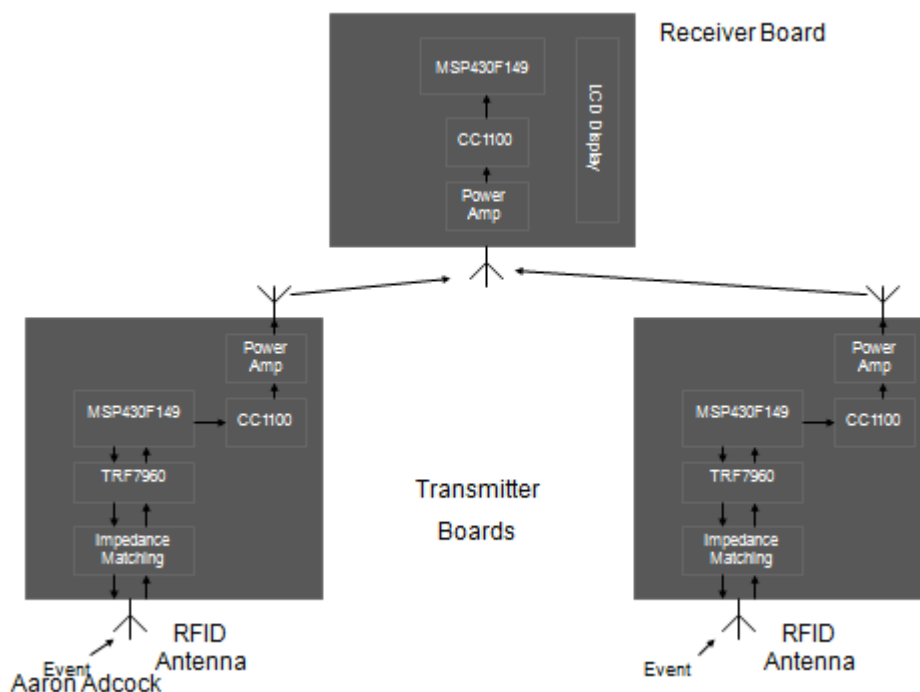


Figure 1- System Diagram

The transmitters and receivers are based around the TI CC1100 sub-1GHz transceiver board. This is a very compact and useful package that will be described in more detail later in the paper. Several CC1100s are connected to custom built microcontroller boards specifically designed for this project.

## CC1100

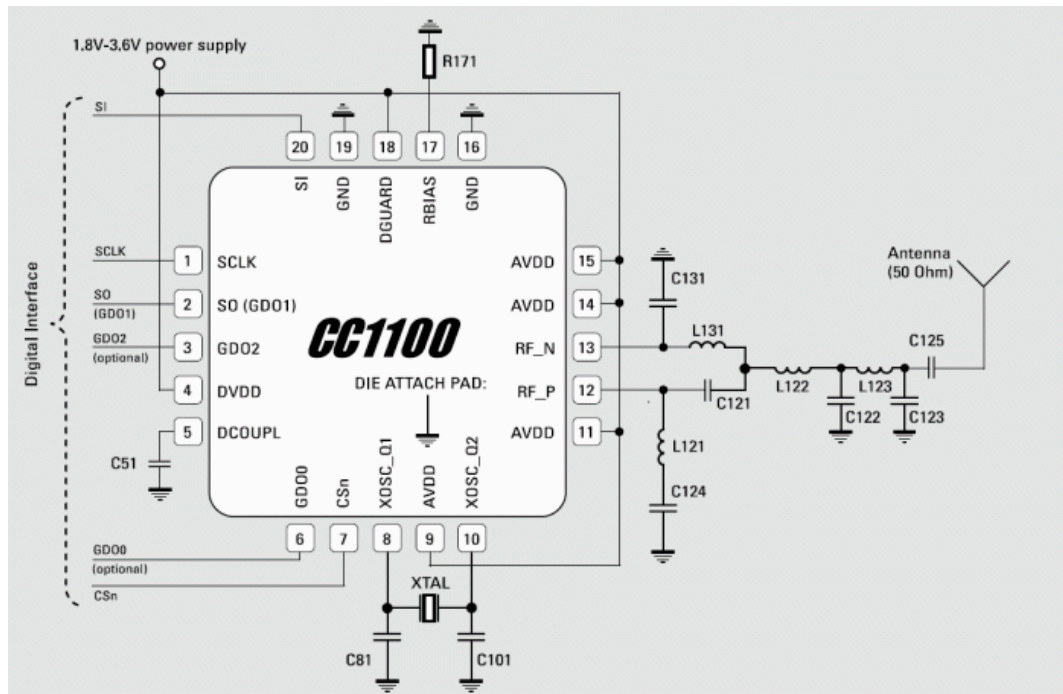


Figure 2-CC1100 [2]

According to the CC100 datasheet, "The CC1100 is a low cost true single chip UHF transceiver designed for very low power wireless applications. The circuit is mainly intended for the ISM (Industrial, Scientific and Medical) and SRD (Short Range Device) frequency bands at 315, 433, 868 and 915MHz, but can easily be programmed for operation at other frequencies in the 300- 348MHz, 400-464MHz and 800-928MHz bands." [1].

## Modulation

The CC1100 supports several different modulation techniques for its RF transmissions.

The supported modulation techniques are:

Table 1: Modulation Techniques

|       |
|-------|
| ASK   |
| OOK   |
| 2-FSK |
| GFSK  |
| MSK   |

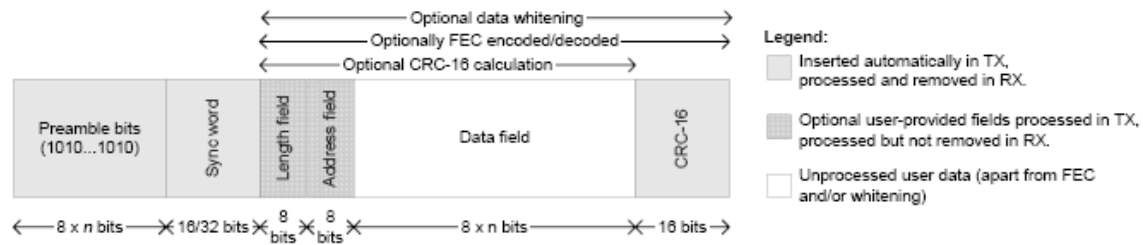
For detailed information about each modulation technique please refer to Appendix A.

Originally, it was planned to allow the receiver board to dynamically switch between multiple modulation techniques to attempt to get a longer range. However, due to time constraints this feature was never implemented. As of the end of the development cycle of this project, the modulation technique has been set to the default MSK, transmitting at a power level defined by the programmer at run time, and at 433 MHz.

## Packet Construction

For purposes of this project, all that is needed is a very simple packet. A packet is the grouping of information to be transmitted. The following figure illustrates the official

packet setup provided by the CC1100:



**Figure 3- Packet Format**

The timer is activated/deactivated when a ping from one of the transmission boards is detected. This allows the receiver board to record the time between two events. The actual packet transmitted is simply full of 1's because the contents are irrelevant. No data is transmitted within the packet, the packet itself is the thing that activates the receiver board timer.

## Programming

The CC1100 is programmed by being connected to a separate MCU. In this case the MCU is the MSP microprocessors contained on the custom built microcontroller boards designed for this project. The CC1100 contains several important registers as well as command strobes that are important to the timing systems' operation.

## Command Strokes

Command strokes are instructions to the CC1100 that control internal sequences of operations within the CC1100. The important command strokes with descriptions are as follows:

**Table 2: Command Strokes**

|       |                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| SRX   | Enable RX. Perform calibration first if coming from IDLE and MCSM0.FS_AUTOCAL=1.                                                                |
| STX   | In IDLE state: Enable TX. Perform calibration first if MCSM0.FS_AUTOCAL=1.If in RX state and CCA is enabled: Only go to TX if channel is clear. |
| SIDLE | Exit RX / TX, turn off frequency synthesizer and exit Wake-On-Radio mode if applicable.                                                         |
| SWOR  | Start automatic RX polling sequence (Wake-on-Radio) as described in Section 19.5 if WORCTRL.RC_PD=0.                                            |

The command strobes allow easy state change of the CC1100 and greatly speed of the programming of the chip. They also insure everything is correctly setup for a given state, and take away a lot of the fallibility that comes inherent with direct programming.

### Power Settings

The CC1100 provides two registers to control the power output level of the transmitter, PATABL and the PA\_POWER bit of FRENDO. PATABL is a special register that can hold up to eight user specified output power settings. The PA\_POWER bit of FRENDO selects which PATABL entry to use. If the current modulation technique requires power ramping, then more than one value should be entered into the PATABL and the highest desired power output entry should be selected with the PA\_POWER bit. Otherwise,

only one PATABLE entry should be entered. If the power output should be changed, simply rewrite the PATABLE with a new single entry. The following figure illustrates the recommended power settings.

| Output Power [dBm] | 433 MHz |                                |
|--------------------|---------|--------------------------------|
|                    | Setting | Current Consumption, Typ. [mA] |
| -30                | 0x04    | 11.5                           |
| -20                | 0x17    | 12.1                           |
| -15                | 0x1C    | 12.7                           |
| -10                | 0x26    | 14.0                           |
| -5                 | 0x57    | 13.7                           |
| 0                  | 0x60    | 15.6                           |
| 5                  | 0x85    | 19.1                           |
| 7                  | 0xC8    | 24.2                           |
| 10                 | 0xC0    | 29.2                           |

**Figure 4- Power Settings**

## The Receiver Board

The receiver is the microcontroller board that has been created to run the timer program.

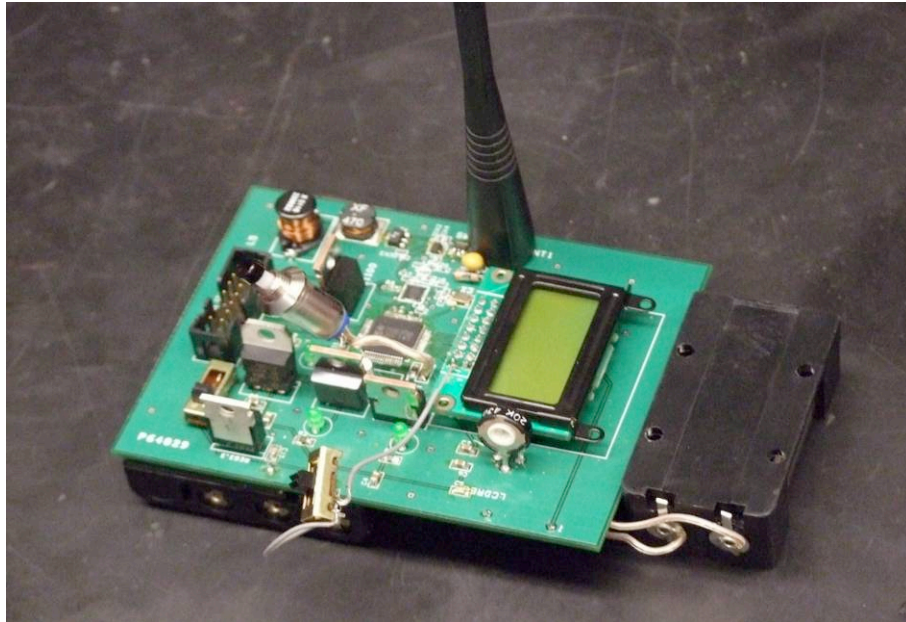
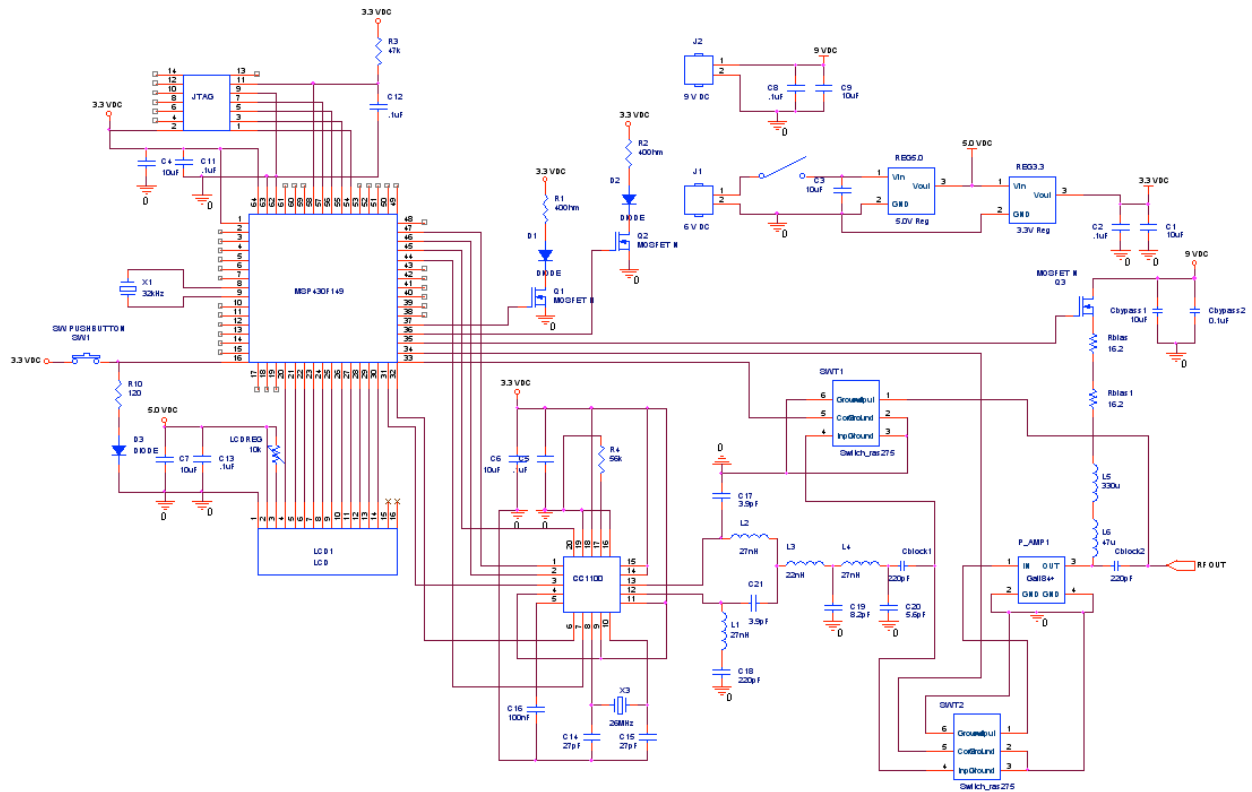


Figure 5- Receiver Board [3]

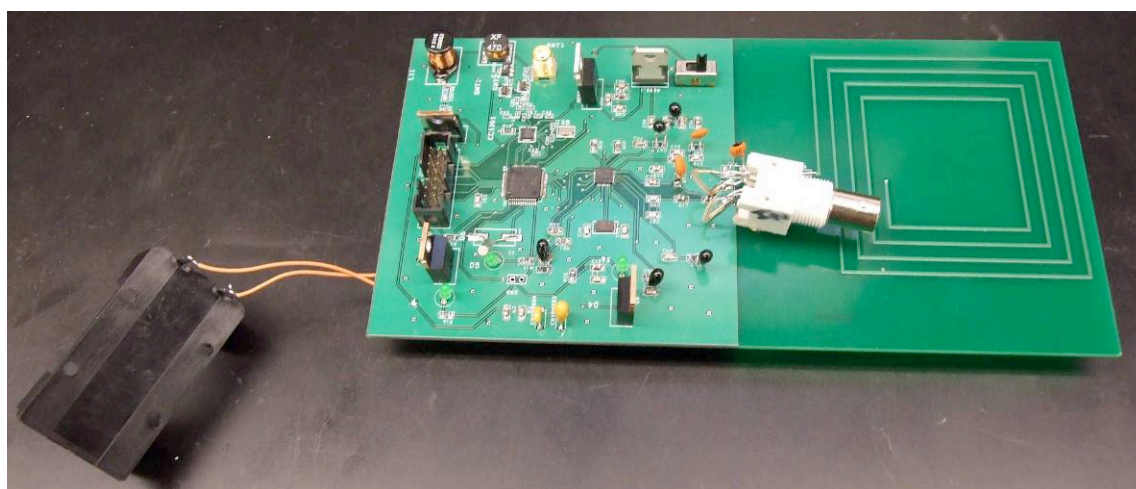
There are several main features of the Receiver Board. It contains a MSP430F149 microprocessor, an LCD screen, and integrated CC1100. The main reason for the LCD screen is to display the current state of the Timer. The board is connected to the computer using a JTAG parallel port -> serial connection, and programmed using the IAR workbench programming suite.

The LCD is made by Crystal Fontz, and is a simple 8 character by 2 lines display. The LCD contains 3 command pins and 8 data pins. The 3 command pins are RS, which controls whether the data sent is a character (High) or a command (Low), R/W, which is always low because low causes data to be sent to the LCD, and R\_EN, which when dropped low causes the LCD to read the data pins. One final note: The LCD was connected to Port 2 and Port 3 on the MSP.

The CC1100 is connected to Port 4 of the MSP, and is interfaced with SPI bit-banging.

### The Transmitter Board

The transmitter board similar to the receiver board, however, it contains a RFID antenna in place of the LCD. It also has an integrated TRF7960 RFID reader. Its main purpose is to interface and transmit data from the sensor so as to trigger the timer on the receiver board. It also contains an interface to connect a power amplifier to increase transmission ranges.



The RFID antenna was designed by Aaron Adcock, and, along with the TRF7960, is designed to work in the 13Mhz – 14Mhz range.

## The Power Amplifier

The following schematic represents the power amplifier chosen to be attached to the transmitter boards.

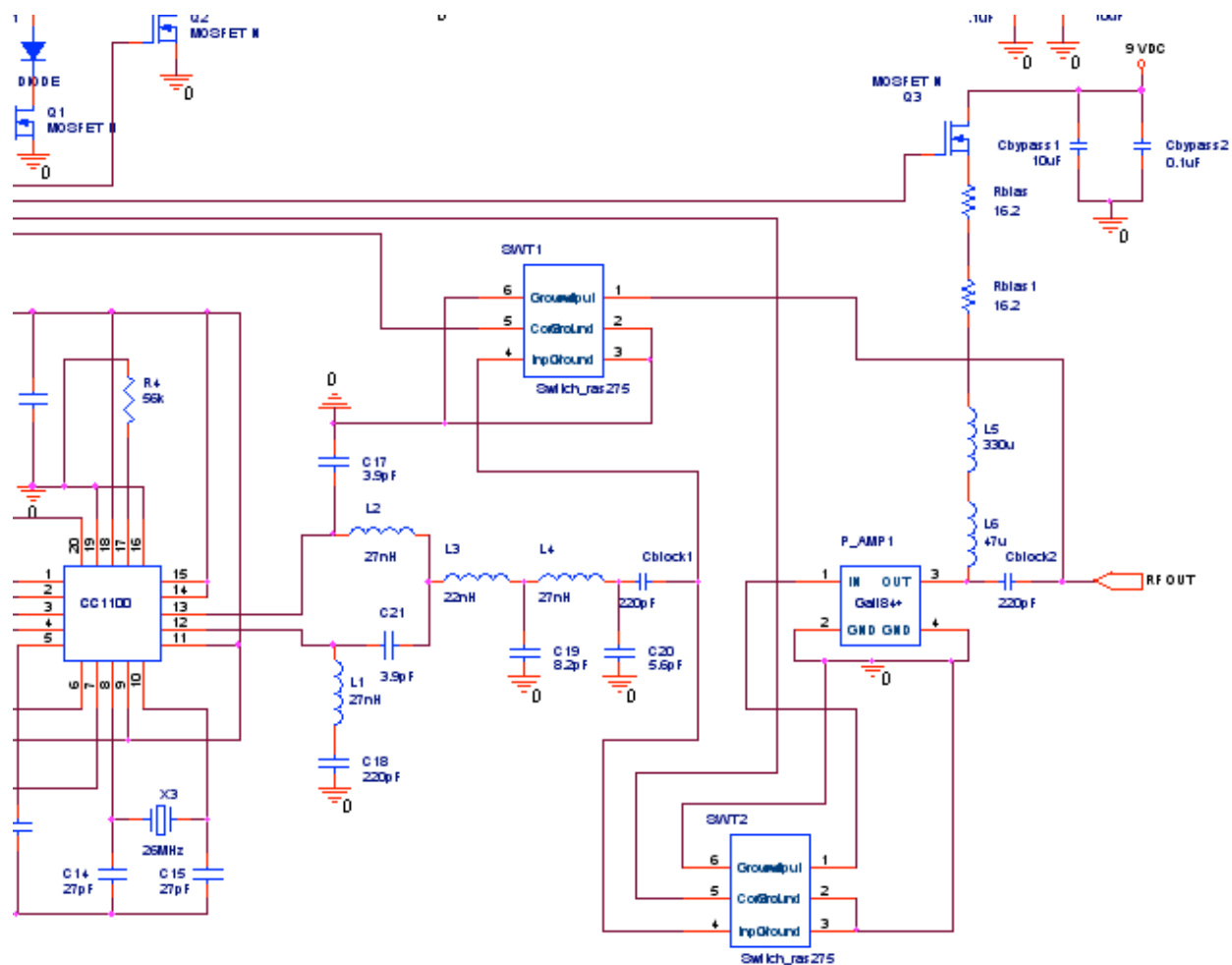


Figure 7- Power Amplifier [5]

The power amplifier is designed to boost the TX output power to allow a greater operational distance. Its maximum output power is 23dBm. There is a software switch inside the transmitter board that must be activated in order to turn on the power amplifier. If this is neglected then transmission will not occur.

## The Software

The programming for the different boards was written in the IAR Workbench programming suite. The code was written in the C programming language, and deals mostly with setting up the registers and SPI on the different boards. The rest of the code mainly deals with the interrupts and the timer program.

### Low Level Programming

#### Registers

Registers are a specific memory location. Therefore, the easiest way to program them is to first define their names as an alias for the memory location via a compile time macro.

For example:

```
#define Register 0x5F
```

#### SPI

Once all of the registers are setup in this fashion, and stored in a header file, you must then setup the SPI, which stands for Serial Peripheral Interface. The steps for interfacing via SPI are illustrated in the following figure:

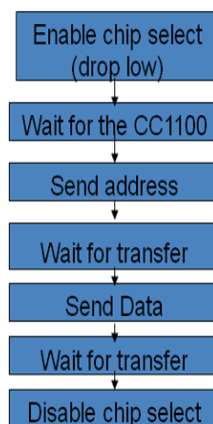


Figure 8- SPI [4]

The SPI was implemented using the bit-banging technique. This means that the whole process is coded manually. An example of bit banging follows [7]:

```

    const unsigned char bitMask8[] = {
    0x80, // binary 10000000
    0x40, // binary 01000000
    0x20, // binary 00100000
    0x10, // binary 00010000
    0x08, // binary 00001000
    0x04, // binary 00000100
    0x02, // binary 00000010
    0x01  // binary 00000001
    };

// This will send data in bit7~0, updating the clock each bit
void send_8bit_serial_data(unsigned char data)
{
    unsigned char x;

    output_high(SD_CS);    // lets select the device

    // Loop through all the bits, 7...0
    for(x = 0; x < 8; x++)
    {
        if(data & bitMask8[x])
        {
            output_high(SD_DI);    // we have a bit, make high
        }
        else
        {
            output_low(SD_DI);    // no bit, make low
        }

        output_low(SD_CLK);    // update clock
        output_high(SD_CLK);    // update clock
    }

    output_low(SD_CS);    // lets DE-select the device
}

```

## LCD

Also, as mentioned earlier, the LCD requires a specific method, which is basically bit-banging, to get commands into its controller. The following code illustrates how to send commands to the LCD.

```

/*
 * Write a command to the LCD.
 */
void LcdWriteCmd(unsigned char cmd)
{
    P2OUT = LCD_RWEN; //E
    P2OUT |= cmd<<3;
    P3OUT = cmd >> 5;
    P3OUT+= POWERAMP;
    P2OUT ^=LCD_RWEN; //
    Delay(100); //waitUntilDone();
} // LcdWriteCmd

```

The command has to be bitwise shifted left by 3 bits to correctly output on port 2 and has to be shifted 5 bits right to be outputted on Port 3. This is because the 5 data pins are on port 2 and 3 are on port 3. LCD\_RWEN is the enable pin and is dropped low to send the command.

## TRF7960

The TRF also requires bitbanging to configure it. The following code is the code used to bitbang the TRF:

```

void bitbangtrf(unsigned char data){
    unsigned char x;

    P3OUT = 0x01;

    for(x = 0; x < 8; x++)
    {
        if(data & bitMask8[x])
        {
            P3OUT |= 0x02;
        }
        else
        {
            P3OUT |= 0x00;;
        }

        P3OUT |= 0x08;
        P3OUT ^= 0x08;
    }

    P3OUT ^= 0x01;
}

```

The TRF requires an initialization routing upon power-up, much like the LCD screen on the receiver board. The following commands must be sent to the TRF to get it to activate:

```
//1st 01 0C 00 03 04 10 00 21 01 13 00 00
//2nd 01 09 00 03 04 F0 00 00 00
//3rd 01 09 00 03 04 F1 FF 00 00
```

These commands were taken from the TRF7960 datasheet. However, after much work, the TRF was still not operating properly. The main reason for this is time restraints as the transmitter board was not populated until 1 day before the project was due.

## The Timer

The Timer is a combination of variable handling and a Timer A interrupt. The main part of the timer is insuring that the variables used to track seconds, minutes, and hours overflow correctly. For example, seconds overflow at 60, minutes overflow at 60, and hours do not overflow. This was implemented using a simple function:

```
void tick(){
    //overflow seconds
    timestring[7]+=1;    //tick seconds

    if(timestring[7]>'9'){ //overflow seconds digit 1
        timestring[7]='0';
        timestring[6]+=1;
    }
    if(timestring[6]=='6'){ //overflow seconds digit 2
        timestring[7]='0';
        timestring[6]='0';
        timestring[4]+=1;
    }
    if(timestring[4]>'9'){ //overflow minutes digit 1
        timestring[4]='0';
        timestring[3]+=1;
    }
    if(timestring[3]=='6'){ //overflow minutes digit 2
        timestring[3]='0';
        timestring[4]='0';
        timestring[1]+=1;
    }
    if(timestring[1]>'9'){ //overflow hours digit 1
        timestring[1]='0';
        timestring[0]+=1;
    }
}
```

The other main part of the timer was the interrupt. It is based off of Timer a:

```

CCTL0 = CCIE;                                // CCR0 interrupt enabled
CCR0 = 32768;
TACTL = TASSEL_1 + MC_1;                     // INCLK, upmode

// Timer A0 interrupt service routine
#pragma vector=TIMER_A0_VECTOR
__interrupt void Timer_A (void)
{

    #ifdef RECEIVER
        lcdhome();
        LcdWriteString("Event 1");
        LcdSecondLine();

        tick();
        char * time = timestring;
        LcdWriteString(time);
    #endif
}

```

The receiver board has a 32,768 Hz crystal, Therefore the CCR0 register was set to the clock speed so that it will overflow every second. The interrupt is set to activate upon CCR0 overflow, so therefore it is a 1 second recurring interrupt. It is used to tick the clock and update the LCD.

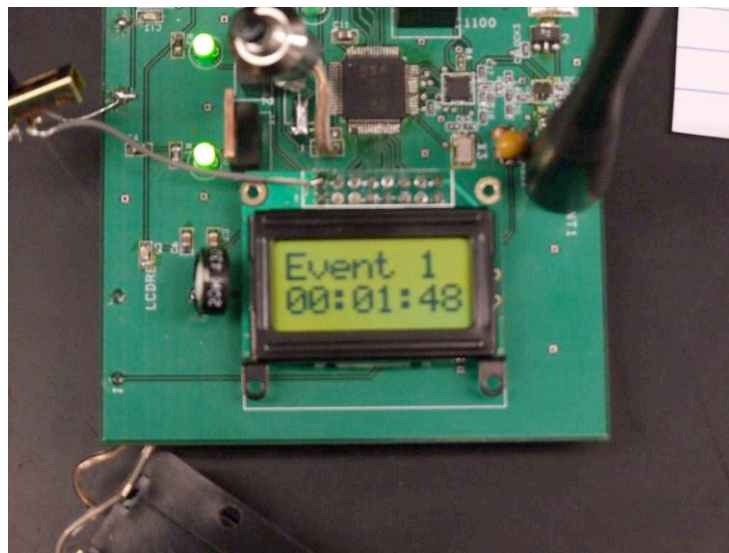


Figure 9 - LCD with Timer

## Testing and Problems

Due to last minute population and programming, some of the testing is not as complete as desired. The main problem areas were the TRF7960, the integrated CC1100 on the receiver board, and the power amplifier.

As previously mentioned, the TRF7960 was not successfully programming. The TRF had several pins of importance, including an enable pin, and a clock pin. The enable pin was basically the power on pin, and the TRF would only work while this pin is high. The clock pin was connected to a clock source. Oscilloscope testing confirmed that the MSP was enabling the correct pins on the TRF, and that its clock source was oscillating. It was unable to be determined why the TRF would not initialize correctly, it was probably a software problem. A similar problem was encountered with the LCD because a delay was not being followed between each command being loaded. However, the TRF datasheet made no mention of a requisite delay.

The integrated CC1100 on the receiver board was another problem area. The GDO0 pin, which is the pin that flags for received data, would never go high. This was probably due to a cold solder joint, but there was insufficient time to correct this problem.

The power amplifier turned out to be another problem area. Initial testing confirmed that it did in fact work to amplify the signal.

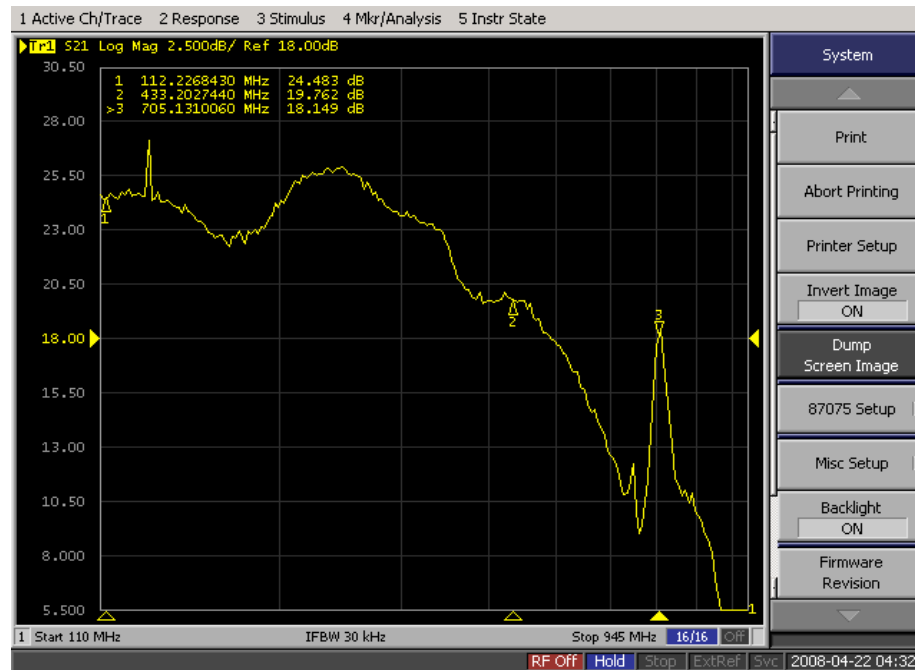


Figure 10- Power Amplifier Power Spectrum [5]

However, upon attempting to use the power amplifier, no transmissions were picked up on the spectrum analyzer. It was suspected that the RF switches were malfunctioning and they were removed. On subsequent removal, and after bypassing the power amplifier, the CC110s on both transmitter boards were functioning correctly.

Another area that was heavily tested before programming commenced was the RFID antenna. As you can see in the next shot, the RFID tags are successfully coupling to the antenna,

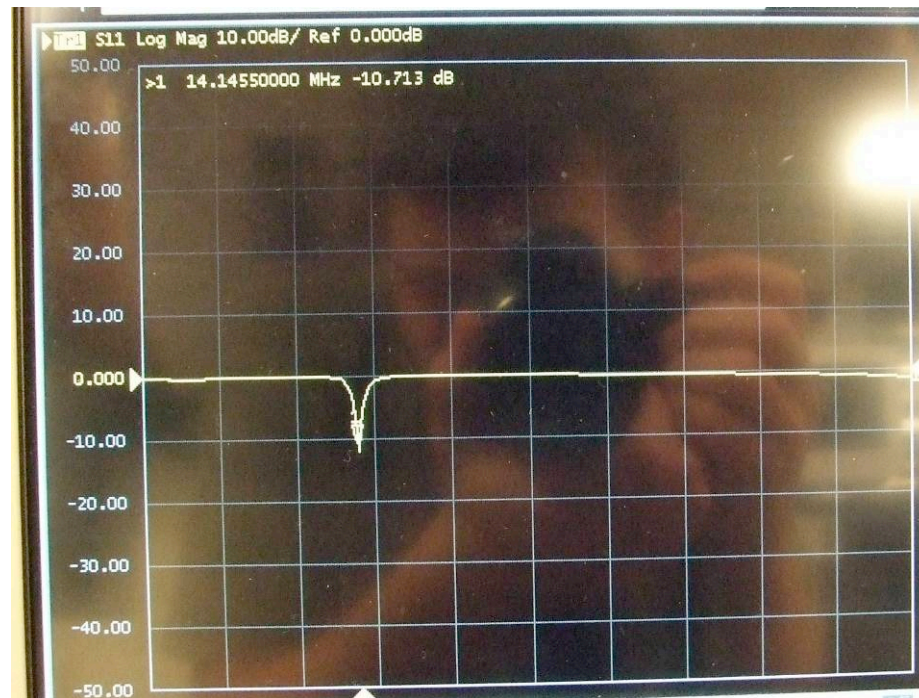


Figure 11- RFID Tag [5]

However, since the TRF7960 was not successfully programmed, the antenna was of no use.

### Uncompleted

Due to time restraints and delays in board design and population, coding was left until the last minute and remains incomplete. The original design for the software is illustrated in the following diagram:

## Software Flow Chart

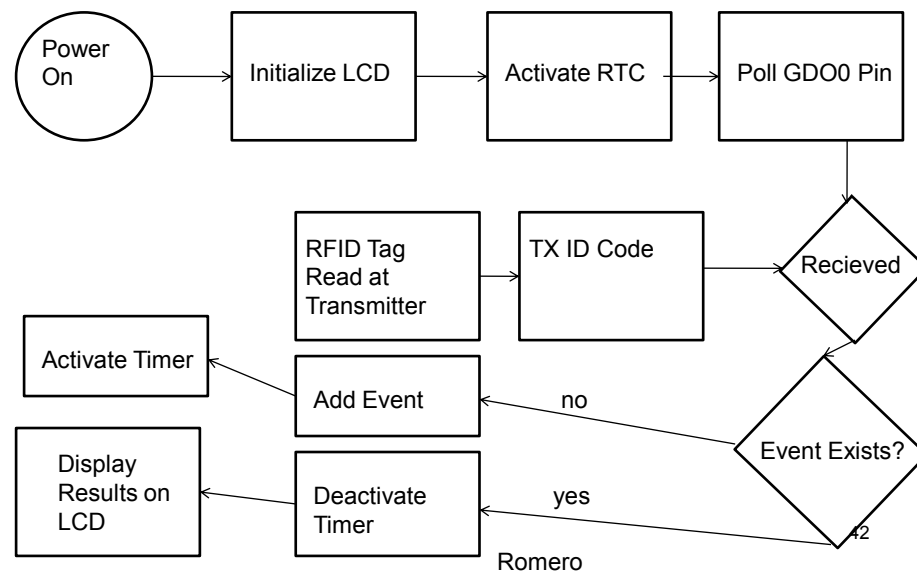


Figure 12 - Software Flow Chart

The TRF7960 and antenna would have been used to read an id number from a RFID tag that is passed over this. The transmitter board would then transmit the id number in a packet to the receiver board. The receiver board would check for the existence of an even with this id number. If there is none, it would activate new event. If it did exist, it would stop the timer for this event. When all events are deactivated, the results will be displayed on the LCD at 5s intervals. This method would have allowed many events to be tracked simultaneously, and would have made full use of the boards' capabilities.

## Conclusion

At the time this project is due, it shows great potential. However, it still has much work to be done to it to make it a functioning system. The two transmitter boards successfully can communicate between each other, which means that the CC1100s are populated and connected correctly. The problems on the receiver board are most likely just due to soldering problems.

The timer works, and is accurate to the second for over an hour in testing conditions, and is successfully displayed on the LCD. If more time were available, the main things that would be corrected are:

- 1.) The TRF7960 must be programmed correctly
- 2.) The receiver board's errors must be corrected, i.e. the CC1100 must work.
- 3.) The RF switches must be replaced and operational for the power amplifier to work.

The main reason why this project fell behind was shifting levels of complexity. The initial project description was quite simple. However, when it was discovered that the preliminary requirements could be completed quickly, new things were added to make the project more interesting and complex. This caused indecision among group members as to how to proceed, and wasted time.

In the end however, everything except the RFID worked, and the project demonstrated that a remote event timer could in fact be implemented, so this project is a partial success.

### **Works Cited**

1. CC1100.pdf, “CC1100 Datasheet”, TI, 3/2/2008
2. <http://www.tdc.co.uk/index.php?key=chipcon>, “TDC.CO.UK, Chipcon Products”
3. Will Gray, “Weekly Presentations” , 4/29/2008
4. Aaron Adcock, “Weekly Presentations”, 4/29/2008
5. Ronjan Mathur, “Weekly Presentations”, 4/29/2008
6. [Http://www.wikipedia.com](http://www.wikipedia.com) , “Wikiedia”, 3/2/2008
7. <http://wikipedia.com/>, bit-banging, 4/29/08

## Appendix A: Pin Outs

MSP430 Pin-out Assignments – Receiver Board

| MSP430 Pin# | MSP430 Port# | Device      | Description   | Device Pin# |
|-------------|--------------|-------------|---------------|-------------|
| 1.          | DVcc         | POWER       | +3.3V         | N/A         |
| 2.          | P6.3         | N/A         | N/A           | N/A         |
| 3.          | P6.4         | N/A         | N/A           | N/A         |
| 4.          | P6.5         | N/A         | N/A           | N/A         |
| 5.          | P6.6         | N/A         | N/A           | N/A         |
| 6.          | P6.7         | N/A         | N/A           | N/A         |
| 7.          | Vref+        | N/A         | N/A           | N/A         |
| 8.          | XIN          | 32KHz XTAL  | XTAL          | XTAL_IN     |
| 9.          | XOUT         | 32KHz XTAL  | XTAL          | XTAL_OUT    |
| 10.         | VeRef+       | N/A         | N/A           | N/A         |
| 11.         | Vref-/VeRef- | N/A         | N/A           | N/A         |
| 12.         | P1.0         | N/A         | N/A           | N/A         |
| 13.         | P1.1         | N/A         | N/A           | N/A         |
| 14.         | P1.2         | N/A         | N/A           | N/A         |
| 15.         | P1.3         | N/A         | N/A           | N/A         |
| 16.         | P1.4         | Pushbutton  | +3.3V input   | N/A         |
| 17.         | P1.5         | N/A         | N/A           | N/A         |
| 18.         | P1.6         | N/A         | N/A           | N/A         |
| 19.         | P1.7         | N/A         | N/A           | N/A         |
| 20.         | P2.0         | LCD         | RS            | LCD_4       |
| 21.         | P2.1         | LCD         | R/W           | LCD_5       |
| 22.         | P2.2         | LCD         | R/W_EN        | LCD_6       |
| 23.         | P2.3         | LCD         | DB0           | LCD_7       |
| 24.         | P2.4         | LCD         | DB1           | LCD_8       |
| 25.         | P2.5         | LCD         | DB2           | LCD_9       |
| 26.         | P2.6         | LCD         | DB3           | LCD_10      |
| 27.         | P2.7         | LCD         | DB4           | LCD_11      |
| 28.         | P3.0         | LCD         | DB5           | LCD_12      |
| 29.         | P3.1         | LCD         | DB6           | LCD_13      |
| 30.         | P3.2         | LCD         | DB7           | LCD_14      |
| 31.         | P3.3         | N/A         | N/A           | N/A         |
| 32.         | P3.4         | N/A         | N/A           | N/A         |
| 33.         | P3.5         | RFSW_1      | P_AMP ON/OFF  | RFSW_1_CTL  |
| 34.         | P3.6         | RFSW_2      | REG_TX ON/OFF | RFSW_2_CTL  |
| 35.         | P3.7         | P_AMP       | Power ON/OFF  | N/A         |
| 36.         | P4.0         | POWER_LED_1 | LED_1 ON/OFF  | N/A         |
| 37.         | P4.1         | POWER_LED_2 | LED_2 ON/OFF  | N/A         |
| 38.         | P4.2         | N/A         | N/A           | N/A         |

|     |          |        |          |           |
|-----|----------|--------|----------|-----------|
| 39. | P4.3     | N/A    | N/A      | N/A       |
| 40. | P4.4     | N/A    | N/A      | N/A       |
| 41. | P4.5     | N/A    | N/A      | N/A       |
| 42. | P4.6     | CC1100 | GDO2     | CC1100_3  |
| 43. | P4.7     | CC1100 | GDO0     | CC1100_6  |
| 44. | P5.0     | CC1100 | CSn      | CC1100_7  |
| 45. | P5.1     | CC1100 | SI       | CC1100_20 |
| 46. | P5.2     | CC1100 | SO/GDO1  | CC1100_2  |
| 47. | P5.3     | CC1100 | SCLK     | CC1100_1  |
| 48. | P5.4     | N/A    | N/A      | N/A       |
| 49. | P5.5     | N/A    | N/A      | N/A       |
| 50. | P5.6     | N/A    | N/A      | N/A       |
| 51. | P5.7     | N/A    | N/A      | N/A       |
| 52. | XT2OUT   | N/A    | N/A      | N/A       |
| 53. | XT2IN    | N/A    | N/A      | N/A       |
| 54. | TDO/TDI  | JTAG   | TDO/TDI  | JTAG_1    |
| 55. | TDI/TCLK | JTAG   | TDI/TCLK | JTAG_3    |
| 56. | TMS      | JTAG   | TMS      | JTAG_5    |
| 57. | TCK      | JTAG   | TCK      | JTAG_7    |
| 58. | RST/NMI  | JTAG   | RST/NMI  | JTAG_11   |
| 59. | P6.0     | N/A    | N/A      | N/A       |
| 60. | P6.1     | N/A    | N/A      | N/A       |
| 61. | P6.2     | N/A    | N/A      | N/A       |
| 62. | AVss     | GND    | GND      | JTAG_9    |
| 63. | DVss     | GND    | GND      | N/A       |
| 64. | AVcc     | POWER  | +3.3V    | N/A       |

## Appendix B: Registers

|      | Write       |         | Read        |            |                                                                        |
|------|-------------|---------|-------------|------------|------------------------------------------------------------------------|
|      | Single byte | Burst   | Single byte | Burst      |                                                                        |
|      | +0x00       | +0x40   | +0x80       | +0xC0      |                                                                        |
| 0x00 |             |         | IOCFG2      |            | R/W configuration registers, burst access possible                     |
| 0x01 |             |         | IOCFG1      |            |                                                                        |
| 0x02 |             |         | IOCFG0      |            |                                                                        |
| 0x03 |             |         | FIFOTHR     |            |                                                                        |
| 0x04 |             |         | SYNC1       |            |                                                                        |
| 0x05 |             |         | SYNC0       |            |                                                                        |
| 0x06 |             |         | PKTLEN      |            |                                                                        |
| 0x07 |             |         | PKTCTRL1    |            |                                                                        |
| 0x08 |             |         | PKTCTRL0    |            |                                                                        |
| 0x09 |             |         | ADDR        |            |                                                                        |
| 0x0A |             |         | CHANNR      |            |                                                                        |
| 0x0B |             |         | FSCTRL1     |            |                                                                        |
| 0x0C |             |         | FSCTRL0     |            |                                                                        |
| 0x0D |             |         | FREQ2       |            |                                                                        |
| 0x0E |             |         | FREQ1       |            |                                                                        |
| 0x0F |             |         | FREQ0       |            |                                                                        |
| 0x10 |             |         | MDMCFG4     |            |                                                                        |
| 0x11 |             |         | MDMCFG3     |            |                                                                        |
| 0x12 |             |         | MDMCFG2     |            |                                                                        |
| 0x13 |             |         | MDMCFG1     |            |                                                                        |
| 0x14 |             |         | MDMCFG0     |            |                                                                        |
| 0x15 |             |         | DEVIATN     |            |                                                                        |
| 0x16 |             |         | MCSM2       |            |                                                                        |
| 0x17 |             |         | MCSM1       |            |                                                                        |
| 0x18 |             |         | MCSM0       |            |                                                                        |
| 0x19 |             |         | FOCCFG      |            |                                                                        |
| 0x1A |             |         | BSCFG       |            |                                                                        |
| 0x1B |             |         | AGCCTRL2    |            |                                                                        |
| 0x1C |             |         | AGCCTRL1    |            |                                                                        |
| 0x1D |             |         | AGCCTRL0    |            |                                                                        |
| 0x1E |             |         | WOREVT1     |            |                                                                        |
| 0x1F |             |         | WOREVT0     |            |                                                                        |
| 0x20 |             |         | WORCTRL     |            |                                                                        |
| 0x21 |             |         | FREND1      |            |                                                                        |
| 0x22 |             |         | FREND0      |            |                                                                        |
| 0x23 |             |         | FSCAL3      |            |                                                                        |
| 0x24 |             |         | FSCAL2      |            |                                                                        |
| 0x25 |             |         | FSCAL1      |            |                                                                        |
| 0x26 |             |         | FSCAL0      |            |                                                                        |
| 0x27 |             |         | RCCTRL1     |            |                                                                        |
| 0x28 |             |         | RCCTRL0     |            |                                                                        |
| 0x29 |             |         | FSTEST      |            |                                                                        |
| 0x2A |             |         | PTEST       |            |                                                                        |
| 0x2B |             |         | AGCTEST     |            |                                                                        |
| 0x2C |             |         | TEST2       |            |                                                                        |
| 0x2D |             |         | TEST1       |            |                                                                        |
| 0x2E |             |         | TEST0       |            |                                                                        |
| 0x2F |             |         |             |            |                                                                        |
| 0x30 | SRES        |         | SRES        | PARTNUM    | Command Strobes, Status registers (read only) and multi byte registers |
| 0x31 | SFSTXON     |         | SFSTXON     | VERSION    |                                                                        |
| 0x32 | SXOFF       |         | SXOFF       | FREQUEST   |                                                                        |
| 0x33 | SCAL        |         | SCAL        | LQI        |                                                                        |
| 0x34 | SRX         |         | SRX         | RSSI       |                                                                        |
| 0x35 | STX         |         | STX         | MARCSSTATE |                                                                        |
| 0x36 | SIDLE       |         | SIDLE       | WORTIME1   |                                                                        |
| 0x37 | SAFC        |         | SAFC        | WORTIME0   |                                                                        |
| 0x38 | SWOR        |         | SWOR        | PKTSTATUS  |                                                                        |
| 0x39 | SPWD        |         | SPWD        | VCO_VC_DAC |                                                                        |
| 0x3A | SFRX        |         | SFRX        | TXBYTES    |                                                                        |
| 0x3B | SFTX        |         | SFTX        | RXBYTES    |                                                                        |
| 0x3C | SWORRST     |         | SWORRST     |            |                                                                        |
| 0x3D | SNOP        |         | SNOP        |            |                                                                        |
| 0x3E | PATABLE     | PATABLE | PATABLE     | PATABLE    |                                                                        |
| 0x3F | TX FIFO     | TX FIFO | RX FIFO     | RX FIFO    |                                                                        |

Table 31: SPI Address Space

## Appendix C-Budget and Gantt

| Estimated Budget (1/18/2008) |                 |                          |                    |            |
|------------------------------|-----------------|--------------------------|--------------------|------------|
| Labor Costs                  |                 |                          |                    |            |
| Name                         | Hourly Pay Rate | Hours Worked             | Total Pay          |            |
| Ronjan Mathur                | \$15.00         | 210                      | \$3,150.00         |            |
| Aaron Adcock                 | \$15.00         | 210                      | \$3,150.00         |            |
| Will Gray                    | \$15.00         | 210                      | \$3,150.00         |            |
| Kyle Romero                  | \$15.00         | 210                      | \$3,150.00         |            |
|                              |                 | Total Labor Cost         | \$12,600.00        |            |
|                              |                 | 75% Overhead             | \$9,450.00         |            |
|                              |                 | Total Labor and Overhead | \$22,050.00        |            |
|                              |                 |                          |                    |            |
| Parts Costs                  |                 |                          |                    |            |
| Part Name                    | Cost            | Quantity                 | Total Cost         |            |
| Misc. Parts                  | \$50.00         | NA                       | \$0.00             |            |
|                              |                 | Total Parts Cost         | \$0.00             |            |
|                              |                 |                          |                    |            |
| Equipment Rental Costs       |                 |                          |                    |            |
| Equipment Name               | Purchase Cost   | Day Rental Rate          | Days Rented        | Total Cost |
| CC1100 x 3                   | \$300.00        | \$0.60                   | 60                 | \$36.00    |
| Development Board            | ?               |                          | 60                 |            |
| Experimenters Board          | \$100.00        | \$0.20                   | 60                 | \$12.00    |
| Oscope                       | \$8,000.00      | \$16.00                  | 60                 | \$960.00   |
|                              |                 |                          | Total Rental Costs | \$1,008.00 |
|                              |                 |                          |                    |            |
| Total Estimated Budget       |                 | \$23,058.00              |                    |            |

| Weekly Budget          |                                                                                     | (4/24/2008)                                  |  |                    |          |            |            |                |
|------------------------|-------------------------------------------------------------------------------------|----------------------------------------------|--|--------------------|----------|------------|------------|----------------|
| Labor Costs            |                                                                                     |                                              |  |                    |          |            |            |                |
| Name                   | Hourly Pay Rate                                                                     | Hours Worked                                 |  | Total Pay          |          |            |            |                |
| Ronjan Mathur          | \$15.00                                                                             | 150                                          |  | \$2,250.00         |          |            |            |                |
| Aaron Adcock           | \$15.00                                                                             | 150                                          |  | \$2,250.00         |          |            |            |                |
| Will Gray              | \$15.00                                                                             | 150                                          |  | \$2,250.00         |          |            |            |                |
| Kyle Romero            | \$15.00                                                                             | 150                                          |  | \$2,250.00         |          |            |            |                |
|                        |                                                                                     | Total Labor Cost                             |  | \$9,000.00         |          |            |            |                |
|                        |                                                                                     | 75% Overhead                                 |  | \$6,750.00         |          |            |            |                |
|                        |                                                                                     | Total Labor and Overhead                     |  | \$15,750.00        |          |            |            |                |
|                        |                                                                                     |                                              |  |                    |          |            |            |                |
| Part Name              | Cost                                                                                | Quantity                                     |  | Total Cost         |          |            |            |                |
| Board Printing         | \$33.00                                                                             | 3                                            |  | \$99.00            |          |            |            |                |
| Transmitter Board      | \$15.77                                                                             | 2                                            |  | \$31.54            |          |            |            |                |
| Receiver Board         | \$22.79                                                                             | 1                                            |  | \$22.79            |          |            |            |                |
| General Board Parts    | \$151.82                                                                            | 1                                            |  | \$151.82           |          |            |            |                |
|                        |                                                                                     | Total Parts Cost                             |  | \$305.15           |          |            |            |                |
|                        |                                                                                     |                                              |  |                    |          |            |            |                |
| Equipment Rental Costs |                                                                                     |                                              |  |                    |          |            |            |                |
| Equipment Name         | Purchase Cost                                                                       | Day Rental Rate                              |  | Days Rented        |          | Total Cost |            |                |
| CC1100 x 3             | \$300.00                                                                            | \$0.60                                       |  | 98                 |          | \$58.80    |            |                |
| Experimenters Board    | \$100.00                                                                            | \$0.20                                       |  | 98                 |          | \$19.60    |            |                |
| Oscope                 | \$8,000.00                                                                          | \$16.00                                      |  | 98                 |          | \$1,568.00 |            |                |
|                        |                                                                                     |                                              |  | Total Rental Costs |          | \$1,646.40 |            |                |
|                        |                                                                                     |                                              |  |                    |          |            |            |                |
| Budget Total           |                                                                                     | \$17,701.55                                  |  |                    |          |            |            |                |
| ID                     |  | Task Name                                    |  | Start              | Finish   | Duration   | % Complete | Resource Names |
| 1                      |  | ☐ Week 1                                     |  | 01/10/08           | 01/16/08 | 5d?        | 100        |                |
| 2                      |  | Gain understanding of the CC1100             |  | 01/10/08           | 01/16/08 | 5d?        | 100        | Mathur         |
| 3                      |  | Research into Packet/Modulation Techniques   |  | 01/10/08           | 01/16/08 | 5d?        | 100        | Adcock         |
| 4                      |  | Look into the Programming Suite (SmartRF)    |  | 01/10/08           | 01/16/08 | 5d?        | 100        | Romero         |
| 5                      |  | Began work with the Experimenter Board       |  | 01/10/08           | 01/16/08 | 5d?        | 100        | Gray           |
| 6                      |  | Researched FCC Bands                         |  | 01/10/08           | 01/10/08 | 1d?        | 100        | Gray           |
| 7                      |                                                                                     | ☐ Week 2                                     |  | 01/17/08           | 01/23/08 | 5d?        | 0          |                |
| 8                      |  | Attempt SmartRF communication with Experi... |  | 01/17/08           | 01/23/08 | 5d?        | 0          | Romero         |
| 9                      |  | Research Packet Creation                     |  | 01/17/08           | 01/23/08 | 5d?        | 0          | Adcock         |
| 10                     |  | Work on Transmitter Board                    |  | 01/17/08           | 01/23/08 | 5d?        | 0          | Gray           |
| 11                     |  | Review Sensor Setups                         |  | 01/17/08           | 01/23/08 | 5d?        | 0          | Mathur         |
| 12                     |                                                                                     | ☐ Week 3                                     |  | 01/24/08           | 01/30/08 | 5d?        | 0          |                |
| 13                     |  | Look into CC1100 Registers / MSP430          |  | 01/24/08           | 01/30/08 | 5d?        | 0          | Romero         |
| 14                     |  | Begin Creating Custom Packets                |  | 01/24/08           | 01/30/08 | 5d?        | 0          | Adcock         |
| 15                     |  | Begin Transmitting with Transmitter Board    |  | 01/24/08           | 01/30/08 | 5d?        | 0          | Gray           |
| 16                     |  | Order Sensors                                |  | 01/24/08           | 01/30/08 | 5d?        | 0          | Mathur         |
| 17                     |                                                                                     | ☐ Week 4                                     |  | 01/31/08           | 02/06/08 | 5d?        | 0          |                |
| 18                     |  | Begin Timing Algorithm                       |  | 01/31/08           | 02/06/08 | 5d?        | 0          | Romero         |
| 19                     |  | Continue Working with Packets                |  | 01/31/08           | 02/06/08 | 5d?        | 0          | Adcock         |
| 20                     |  | Begin working with MSP/Sensor Interface      |  | 01/31/08           | 02/06/08 | 5d?        | 0          | Gray           |
| 21                     |  | Testing of Sensors                           |  | 01/31/08           | 02/06/08 | 5d?        | 0          | Mathur         |

| ID | Task Name                                     | Start           | Finish          | Duration   | % Complete | Resource Names |
|----|-----------------------------------------------|-----------------|-----------------|------------|------------|----------------|
| 22 | <b>Week 5</b>                                 | <b>02/07/08</b> | <b>02/13/08</b> | <b>5d?</b> | <b>0</b>   |                |
| 23 | Research Sensor ID # Techniques               | 02/07/08        | 02/13/08        | 5d?        | 0          | Romero         |
| 24 | Testing Sensors with Transmission             | 02/07/08        | 02/13/08        | 5d?        | 0          | Mathur         |
| 25 | Transfer Sensor Data to packets               | 02/07/08        | 02/13/08        | 5d?        | 0          | Gray           |
| 26 | Finalize Packet Setup                         | 02/07/08        | 02/13/08        | 5d?        | 0          | Adcock         |
| 27 | <b>Week 6</b>                                 | <b>02/14/08</b> | <b>02/20/08</b> | <b>5d</b>  | <b>0</b>   |                |
| 28 | Begin testing packet Data                     | 02/14/08        | 02/20/08        | 5d         | 0          | Adcock         |
| 29 | Begin Building Sensor/Transmitter Package     | 02/14/08        | 02/20/08        | 5d         | 0          | Mathur         |
| 30 | Begin Interpreting Packet Data                | 02/14/08        | 02/20/08        | 5d         | 0          | Romero         |
| 31 | Begin Power algorithm                         | 02/14/08        | 02/20/08        | 5d         | 0          | Gray           |
| 32 | <b>Week 7</b>                                 | <b>02/21/08</b> | <b>02/27/08</b> | <b>5d</b>  | <b>0</b>   |                |
| 33 | Begin Modulation Tests                        | 02/21/08        | 02/27/08        | 5d         | 0          | Adcock         |
| 34 | Continue Building Sensor/Transmitter Package  | 02/21/08        | 02/27/08        | 5d         | 0          | Mathur         |
| 35 | Continue interpreting packet data             | 02/21/08        | 02/27/08        | 5d         | 0          | Romero         |
| 36 | Continue Power Algorithm                      | 02/21/08        | 02/27/08        | 5d         | 0          | Gray           |
| 37 | <b>Week 8</b>                                 | <b>02/28/08</b> | <b>03/05/08</b> | <b>5d</b>  | <b>0</b>   |                |
| 38 | Decide Modulation method                      | 02/28/08        | 03/05/08        | 5d         | 0          | Adcock         |
| 39 | Finish Sensor/Transmitter Package             | 02/28/08        | 03/05/08        | 5d         | 0          | Mathur         |
| 40 | Finish Timing Algorithm                       | 02/28/08        | 03/05/08        | 5d         | 0          | Romero         |
| 41 | Finish Power Algorithm                        | 02/28/08        | 03/05/08        | 5d         | 0          | Gray           |
| 42 | <b>Week 9</b>                                 | <b>03/06/08</b> | <b>03/12/08</b> | <b>5d</b>  | <b>0</b>   |                |
| 43 | Test Packet Transmission (Separate)           | 03/06/08        | 03/12/08        | 5d         | 0          | Adcock         |
| 44 | Test Power Algorithm (Separate)               | 03/06/08        | 03/12/08        | 5d         | 0          | Gray           |
| 45 | Test Timing Algorithm (Separate)              | 03/06/08        | 03/12/08        | 5d         | 0          | Romero         |
| 46 | Design Full System Test                       | 03/06/08        | 03/12/08        | 5d         | 0          | Mathur         |
| 47 | <b>Week 10</b>                                | <b>03/13/08</b> | <b>03/26/08</b> | <b>10d</b> | <b>0</b>   |                |
| 48 | Test Packet Transmission (Full System)        | 03/20/08        | 03/26/08        | 5d         | 0          | Adcock         |
| 49 | Test Power Algorithm (Full System)            | 03/20/08        | 03/26/08        | 5d         | 0          | Gray           |
| 50 | Test Timing/Receiver Algorithms (Full System) | 03/20/08        | 03/26/08        | 5d         | 0          | Romero         |
| 51 | Test Sensors (Full System)                    | 03/20/08        | 03/26/08        | 5d         | 0          | Mathur         |
| 52 | <b>Week 11</b>                                | <b>03/13/08</b> | <b>03/19/08</b> | <b>5d</b>  | <b>0</b>   |                |
| 53 | Spring Break!                                 | 03/13/08        | 03/19/08        | 5d         | 0          | All            |
| 54 | <b>Week 12</b>                                | <b>03/27/08</b> | <b>04/02/08</b> | <b>5d</b>  | <b>0</b>   |                |
| 55 | Fix Packet Problems                           | 03/27/08        | 04/02/08        | 5d         | 0          |                |
| 56 | Fix Power Problems                            | 03/27/08        | 04/02/08        | 5d         | 0          |                |
| 57 | Fix Receiver Problems                         | 03/27/08        | 04/02/08        | 5d         | 0          |                |
| 58 | Fix Sensor Problems                           | 03/27/08        | 04/02/08        | 5d         | 0          |                |
| 59 | <b>Week 13</b>                                | <b>04/03/08</b> | <b>04/09/08</b> | <b>5d</b>  | <b>0</b>   |                |
| 60 | Continue System Testing                       | 04/03/08        | 04/09/08        | 5d         | 0          | Adcock         |
| 61 | Continue System Testing                       | 04/03/08        | 04/09/08        | 5d         | 0          | Gray           |
| 62 | Continue System Testing                       | 04/03/08        | 04/09/08        | 5d         | 0          | Romero         |
| 63 | Continue System Testing                       | 04/03/08        | 04/09/08        | 5d         | 0          | Mathur         |
| 64 | <b>Week 14</b>                                | <b>04/10/08</b> | <b>04/16/08</b> | <b>5d</b>  | <b>0</b>   |                |
| 65 | Prepare Demonstration                         | 04/10/08        | 04/16/08        | 5d         | 0          | Romero         |
| 66 | Make Poster For Demonstration                 | 04/10/08        | 04/16/08        | 5d         | 0          | Gray           |
| 67 | Work out transmission errors                  | 04/10/08        | 04/16/08        | 5d         | 0          | Adcock         |
| 68 | Work out Mechanical errors                    | 04/10/08        | 04/16/08        | 5d         | 0          | Mathur         |
| 69 | <b>Week 15</b>                                | <b>04/17/08</b> | <b>04/23/08</b> | <b>5d</b>  | <b>0</b>   | Mathur         |
| 70 | Demo Day!                                     | 04/17/08        | 04/23/08        | 5d         | 0          | All            |
| 71 | <b>Week 16</b>                                | <b>04/24/08</b> | <b>04/30/08</b> | <b>5d</b>  | <b>0</b>   |                |
| 72 | Final Presentation/Report                     | 04/24/08        | 04/30/08        | 5d         | 0          | All            |